Toka Alokda, M.Sc. (she/they)
Ph.D. researcher in Cosmology
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Research interests

- **Cosmology:** Large-scale structure; structure formation; statistical analysis; theoretical modeling; numerical simulations; primordial non-Gaussianity.
- **Statistics:** Bayesian inference; information geometry; uncertainty quantification in analysis pipelines.
- **Machine learning:** Neural-network architecture, optimization, and validation; neural simulation-based inference (SBI); interpretable machine learning for science.
- **Scientific computing:** GPU-accelerated code development; differentiable programming; scalable scientific software and applications.

Education

University of Bonn (Rheinische Friedrich-Wilhelms-Universität Bonn) - Bonn, DE.
Ph.D. - Astrophysics
International Max Planck Research School for Astronomy and Astrophysics (IMPRS A&A).
Advisor(s): Prof. Dr. Cristiano Porciani, Dr. Alexander Eggemeier.
Topic: Exploring Primordial Non-Gaussianity through the Large Scale Structure with Simulations and Machine Learning. **June '24 - Present**

University of Bonn (Rheinische Friedrich-Wilhelms-Universität Bonn) - Bonn, DE.
M.Sc. - Astrophysics.
Bonn-Cologne Graduate School for Physics and Astronomy (BCGS).
Advisor(s): Prof. Dr. Cristiano Porciani.
Thesis: "Identification of Protohalos with Deep Learning" (Grade: 1.0[1])
Grade: 1.6 (ranking: 3rd in the class of 2023). **Oct. '21 - Dec. '23**

University of Science and Technology at Zewail City - Giza, EG.
B.Sc. - Physics of Earth and Universe; Astrophysics concentration.
Advisor(s): Prof. Dr. Shaaban Khalil.
Thesis: "A Dark Matter Explanation for Galactic Positron Flux" (Grade: 4.0/4.0)
GPA: 3.63 [2] (Cum Laude - final semester: 4.0/4.0) **Sep. '17 – Jul. '21**

Academic appointments

DFG Cluster of Excellence DYNAVERSE - PhD researcher. **Apr. '26 - Present**
Argelander-Institut für Astronomie, University of Bonn - PhD researcher. **Jun. '24 - Present**
University of Bonn - Astrophysics MSc program - Graduate teaching assistant. **Oct. '22 - Feb. '24**

[1] Indicated in the German system; whereas 1.0 is the highest grade and 5.0 is the lowest.
[2] Indicated in the American system; whereas 4.0 is the highest grade and 0.0 is the lowest.

Awards & Honors

COSMOVERSE COST action - travel grant **May '26**
Les Houches School of Physics - travel grant **Jul. '25**
G2net COST action - travel grant **Mar. '23**
BCGS honors scholarship - Full merit-based scholarship and stipend to pursue a Master's in Astrophysics in Bonn. **Oct. '21 - Sep. '23**
Provost's Honor Roll - UST Zewail City (semesterly academic achievement) **Jul. '21**
UST Zewail City scholarship - Full merit-based scholarship. **Sep. '17 - Jul. '21**
MFK excellence scholarship (top 100 students) - Declined. **Jul. '17**
Egyptian national secondary certificate (Thanaweya Amma) distinction (top 50 students). **Jul. '17**

Publications

Journal articles

Alokda, T., Porciani, C. and Eggemeier, A. (2026). *Coverage is not enough: Frequentist tests of simulation-based inference for primordial non-Gaussianity.* A&A, 714, A66 (2026). DOI:10.1051/0004-6361/202660668

Alokda, T. and Porciani, C. (2025). *Segmenting proto-halos with vision transformers.* JCAP 11 (2025) 083. DOI:10.1088/1475-7516/2025/11/083

Other scholarly contributions

Alokda, T. (2026). Invited referee report on *"Bridging perturbation theory and simulations: Initial conditions and fast integrators"* by O. Hahn. SciPost Physics Lecture Notes. DOI:10.21468/SciPost.Report.12637

Manuscripts in preparation

Alokda, T. with Porciani, C. and Eggemeier, A. "Taming the *n*-point hierarchy: Wavelet polyspectra for primordial non-Gaussianity" (working title)

Alokda, T. with Porciani, C., Eggemeier, A. and Ayromlou, M. "Galaxy-selection dependence of the clustering response to local primordial non-Gaussianity in L-Galaxies" (working title)

Selected talks and presentations

"Coverage is not enough: What SBI posteriors of f_NL are actually telling you"
Munich, DE - Contributed talk in the 2026 AG meeting. **Sep. '26**

"Segmenting Protohalos with Vision Transformers"
Munich, DE - Contributed talk in the 2026 AG meeting. **Sep. '26**

"Coverage is not enough: Frequentist test of simulation-based inference for primordial non-Gaussianity"
A Coruna, ES - Contributed poster presentation and lightning talk in the "New frontiers in Cosmology" conference. **Aug. '26**

"Coverage is not enough: What SBI posteriors of f_NL are actually telling you"
Waterloo, CA - Contributed talk in the "Advancing simulation-based and field-level inference for Cosmology" conference. **Jun. '26**

"The quest for primordial non-Gaussianity: Opportunities and Challenges"
Sofia, BG - Contributed lightning talk at the CosmoVerse 2026 workshop. **May '26**

"Segmenting Protohalos with Vision Transformers"
Cologne, DE - Contributed talk in the Astroinformatics 2026 conference. **Feb. '26**

"Identification of Protohalos with Deep Learning"

Les Houches, FR - Contributed talk at the Les Houches - Dark Universe school. **Jun. '25**
"Identification of Protohalos with Deep Learning"
Cambridge, MA - Online poster presentation in the first AstroAI workshop. **Jun. '24**

Teaching Experience

University of Bonn - M.Sc. of Astrophysics program
1. The large scale structure as revealed by redshift surveys (Tutorial) - 1 semester
2. Maths for Theoretical Astrophysics and Cosmology (Support Lecture) - 4 semesters
3. Advanced Topics in Cosmology (Tutorial) - 2 semesters
4. Astrophysics of Galaxies (Tutorial) - 1 semester

University of Science and Technology at Zewail City - B.Sc. of Physics program
1. Quantum Mechanics I (Tutorial) - 1 semester.

Technical Skills

Programming & High-Performance Computing
- Languages: Python (Advanced), C++/CUDA C++ (Advanced), Julia (Intermediate), MATLAB (Intermediate), Mathematica (Intermediate), Fortran (Basic).
- HPC & Systems: Linux/UNIX, Bash, Slurm Workload Manager, Parallel Computing.

Scientific Computing & Machine Learning
- Frameworks & Libraries: PyTorch, TensorFlow, JAX, Keras.
- Methods: Differentiable Programming, Physics-Informed ML, Scientific ML.

Astrophysics & Cosmology Software
- Simulation & Halo Tools: GADGET-4, L-Galaxies, AHF, PINOCCHIO.
- Analysis & Cosmology: CLASS, Pylians, PySpectrum, Nbodykit.

Development & Documentation: Git, LaTeX, MS Office.

Language Skills

Arabic (native) - English (C2) - German (B1) - French (A2)

Service and outreach

Serving as a supervisor in the Astro seminar course at the AIfA, Bonn **Oct. '26 - Present**
Science communication with self-made Astro web-comics on Instagram **Sep. '26 - Present**
DYNAVERSE postdoc selection committee **Jul. '26**
"Taking Up Space: Life from the PhD trenches" - Online blog column **Apr. '26 - Present**
AIfA science day - organizer **Apr. '26**
Member of the AIfA DEI committee - organizer in several social events **Jul. '24 - Sep. '25**
"Tangled in the Web: The rise of AI in Astronomy" - Public talk in AoT Bonn **Oct. '24**
Unbiased cosmology student seminar - Organizer and blog writer **Oct. '22 - Sep. '24**
Zewail City Open Courseware - Note-taking and social media management **Oct. '18 - Jul. '21**

I also contributed to several scientific initiatives for the Egyptian public within 2018-2021. Highlights include the Zewail City annual science festival, TEDx Zewail City, Zewail City kids' science camp, and a locally-organized summer training in basic Mathematics for school girls.

References are available upon request